

AI CAREERS FOR WOMEN LINKEDIN FELLOWSHIP

[Academic Resource Optimization Dashboard]

Project Domain

Excel

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:



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Year - 2026

ACKNOWLEDGEMENT

We want to sincerely thank everyone who supported us throughout this project. A special thanks goes to our mentor, Umamaheswari , for their constant guidance and belief in our work. Their feedback turned our challenges into learning opportunities and kept us motivated from start to finish.

We are also incredibly grateful to the **Edunet Foundation, Microsoft, LinkedIn, and SAP** for this fellowship. It provided us with the tools and platform to grow our skills in AI and Data Analytics. We also appreciate **National Post Graduate College** for giving us the space and resources to work. Lastly, a big thank you to our family and friends—your encouragement made the long hours much easier.

ABSTRACT OF THE PROJECT

This project is part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP. The fellowship is designed to equip students with practical, hands-on skills in Artificial Intelligence, Data Analytics, and Business Intelligence tools.

Our project, **Academic Resource Optimization Analysis**, addresses the challenge of identifying students at risk due to poor attendance and a lack of digital infrastructure. In many educational settings, raw data is often poorly structured or contains "Text-to-Number" formatting errors that prevent accurate analysis. To solve this, we expanded an initial 30-row dataset to a comprehensive 500-student database to better simulate real-world trends.

The methodology involved rigorous data cleaning to resolve "Red Syntax" errors in Column D (Marks), ensuring all data was numerically valid for complex calculations. Using advanced Excel features like **Nested IF** and **IFS** logic, we engineered a "Resource Need Level" index to categorize students as "Critical," "High," "Medium," or "Low".

Key results achieved through the use of **Pivot Tables** include the discovery of a direct 40% performance gap between high and low-attendance students and the identification of Mathematics as the subject requiring the most remedial intervention. We also leveraged **AI Logic Design** to debug formula bottlenecks and ensure the "Top 50" topper list scaled correctly with 500 records.

In conclusion, this project demonstrates how data cleaning and logical feature engineering can transform "noisy" administrative records into a powerful decision-support system, enabling schools to distribute resources more effectively based on student need.

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CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project. The fellowship projects span multiple domains including Data Analytics, Business Intelligence, Healthcare Analytics, Content Generation, and AI-assisted Reporting.

1.1 Problem Statement

The core challenge of this project is the **"Analytical Gap"** in educational administration, where vast amounts of student data remain underutilized due to poor data integrity and non-computable formats (e.g., numbers stored as text).

- **Significance:** This leads to **"Delayed Intervention."** Without clean data, administrators cannot identify at-risk students—such as those failing due to lack of internet access—until it is too late, resulting in higher dropout rates and inefficient resource allocation.
- **Target Users:** This framework is designed for **School Administrators, Academic Counselors, and Educational NGOs** who require data-driven evidence to distribute tuition support and test-prep resources.
- **Limitations of Existing Approaches:** Current manual tracking methods lack **automated logical branching** and fail to scale effectively. As data grows from 30 to 500 records, manual oversight becomes impossible. Existing tools also struggle with "Type Mismatches" (Red Syntax errors), which this project resolves by building a scalable, automated cleaning and categorization pipeline.

1.2 Motivation

The core motivation for this project is to leverage data as a tool for educational equity. By identifying students who lack digital access or consistent attendance, we can transform disorganized records into proactive support systems.

- **Personal & Professional Significance:** This project provided hands-on experience in high-level data engineering, specifically in scaling a dataset to 500 records and resolving complex "Type Mismatch" errors. It emphasizes the critical role of data cleaning in the analytics lifecycle.
- **Real-World Impact:** This framework enables schools to automatically flag at-risk students, optimize limited budgets for tuition and test-prep based on a "Resource Need Level," and identify subject-specific performance gaps in real-time.
- **Industry Alignment:** The project reflects the shift toward "Data-Driven Decision Making" and "Logic-Based Categorization." By utilizing AI-assisted logic design to build complex

formulas, it demonstrates how modern analysts use AI to solve technical bottlenecks and accelerate workflows.

1.3 Objectives

1. **Data Exploration:** To collect and audit a dataset of 500 student records, identifying structural inconsistencies and data integrity issues.
2. **Data Normalization:** To sanitize raw data by correcting "Type Mismatch" errors (converting numbers stored as text) and standardizing fields to ensure computational accuracy.
3. **Advanced Feature Engineering:** To develop a multi-conditional prioritization engine using Nested IF and IFS logic to categorize students based on academic performance and digital infrastructure access.
4. **Multi-Dimensional Analysis:** To utilize Pivot Tables to discover non-obvious correlations between student attendance, subject-specific failure rates, and resource availability.
5. **Interactive Reporting:** To construct a dynamic "Topper Leaderboard" and performance dashboard that utilizes Value Filtering to manage large-scale data visualization effectively.
6. **AI Integration:** To leverage AI as a Logic Architect for debugging complex categorical formulas and automating the generation of a statistically significant 500-row synthetic dataset.
7. **Project-Specific Objective:** To design an Academic Resource Optimization Dashboard in Microsoft Excel that analyzes student performance indicators such as subject-wise scores, grade distribution, attendance impact, and pass–fail trends to support data-driven academic decision-making.

1.4 Scope of the Project

The scope of this project is defined by the following boundaries and technical constraints:

- **Data Coverage:** This project covers a synthetic dataset of 500 student records, encompassing 13 distinct variables including academic marks, attendance percentages, and socio-economic indicators like internet access and tuition status.
- **Technical Stack:** The implementation is strictly limited to Microsoft Excel for data processing and AI Chatbot/Copilot for logic consultation and formula optimization.
- **Analytical Focus:** The project primarily focuses on Data Normalization (resolving type mismatches), Logical Feature Engineering (creating "Resource Need" categories), and Diagnostic Visualization through Pivot Tables and Leaderboards.

- Exclusions: The project does not include real-time database connectivity, live API integrations, or predictive Machine Learning models. It is focused on descriptive and diagnostic analytics rather than predictive forecasting.
- Project Intent: This project is intended as an AICW Fellowship showcase to demonstrate proficiency in data handling and AI-assisted logic design; it is not a production-level school management system.

CHAPTER 2

Literature Survey

This chapter reviews existing research, methodologies, and technical tools relevant to improving student achievement through data-driven insights and spreadsheet technology. It identifies the foundational theories and technological frameworks that support the development of the **Student Resource Optimization Dashboard**.

2.1 Existing Work and Research

- **Impact of Spreadsheet Technology on Achievement:** Research indicates that utilizing Microsoft Excel in educational settings significantly improves student achievement and retention in complex mathematical concepts, such as quadratic functions. Studies have shown that students taught using Excel-based pedagogies outperform those taught through conventional, teacher-centered methods.
- **Active Learning and Conceptualization:** Literature suggests that spreadsheet tools allow students to interactively model and simulate mathematical situations. This practical, exploratory approach helps students conceptualize relationships between numerical, graphical, and algebraic representations more effectively than mechanical memorization.
- **The "Digital Divide" and Technology Access:** Significant differences in student performance have been observed between schools with high technology implementation and those without. Research confirms that integrating ICT as a teaching strategy is essential for closing achievement gaps and preparing students for a techno-cultural global society.
- **Theoretical Foundations:**
 - **Thorndike's Decay Theory:** Highlighting that extensive practice—facilitated by Excel's rapid feedback—promotes the encoding of information for long-term retention.
 - **Vygotsky's Socio-Cultural Theory:** Reorienting learning from an individualistic task to a tool-mediated process.

Reference link [\(PDF\) Utilizing microsoft excel spreadsheet technology to improve senior secondary school students' achievement and retention in quadratic function in Nigeria education](#)

2.2 Tools and Technologies Review

- **Microsoft Excel:** A primary tool for data cleaning, transformation, and visualization. Key advantages include its utility as a general-purpose tool, its ubiquity across educational and business environments, and its user-friendly interface that provides immediate feedback for hypothesis testing.
- **Data Generation & Scaling:** Synthetic data generation techniques, such as the =RANDBETWEEN and CHOOSE functions, allow for the creation of robust, large-scale datasets from limited samples to facilitate deep analysis.
- **AI Logic Architecture (Google Gemini/Microsoft Copilot):** Generative AI tools serve as "Logic Architects," enabling the creation of complex nested formulas (e.g., =IFS combined with AND) to automate student categorization based on multiple risk factors.
- **Pivot Tables and Dashboards:** These Excel features allow for the segmentation of databases into actionable insights, such as correlating internet access directly with academic "Critical" status.

2.3 Gaps Identified

- **Reliance on Memorization:** Traditional teaching methods often lead students to focus on algebraic operations and memorized formulas rather than building a functional understanding of data relationships.
- **Limited Integration of AI in Student Analytics:** While complex machine learning models exist, there is a lack of documentation on using accessible AI tools like Copilot or Gemini to assist non-technical users in building domain-specific optimization dashboards.
- **Underutilization of Immediate Feedback:** Many educational approaches fail to leverage the "immediate feedback" loop provided by spreadsheet technologies, which is proven to increase student engagement and problem-solving skills.
- **Standardized Frameworks for AI Prompting:** There is an ongoing need to develop standardized prompt libraries for educational resource management to reduce human error and accelerate data analysis.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

The system employs a modular "Ingestion-to-Insight" architecture, ensuring raw data is validated before being transformed into actionable intelligence.

- **Data Source / Input:** A scaled dataset of 500 student records containing 13 variables, including academic marks, attendance, and digital infrastructure status.
- **Data Cleaning (Validation Layer):** This "Normalization Layer" resolved "Data Impedance Mismatches" by converting text-based marks into integers using the VALUE() function, standardizing categorical "Yes/No" labels, and handling null values to ensure computational stability.
- **Analysis / Processing (Logic Engine):** A Multi-Tiered Logic Engine was implemented using AI-architected Nested IF/IFS formulas. This layer engineered new features, such as the "Resource Need Level," by cross-referencing performance against infrastructural gaps.
- **Visualization / Output:** The final output is an Interactive Diagnostic Dashboard featuring a "Top 50" Leaderboard, Pivot Charts correlating attendance to performance, and summary tables identifying high-priority subjects.

3.2 Modules / Components Used

The project is structured into five core functional modules that work together to transform raw student data into administrative insights:

- **Module 1 — Data Ingestion & Scaling:** This component involved importing the initial student data and utilizing the RANDBETWEEN and VLOOKUP functions to scale the dataset to 500 unique records, ensuring a statistically significant sample for analysis.
- **Module 2 — Data Validation & Cleaning:** This module acted as a "Quality Gate." It focused on resolving Type Mismatches (converting text-stored marks to numerical values), removing duplicate entries, and standardizing categorical data (e.g., Internet Access and Tuition status) to prevent formula errors.
- **Module 3 — Logic & Feature Engineering:** The "Engine" of the project. This module used complex Nested IF and IFS calculations to create new data points, specifically the "Resource Need Level" and "Student Status" (e.g., Critical, Stable, or Excellent).
- **Module 4 — Reporting & Visualization:** This component used Pivot Tables and Pivot Charts to aggregate the 500 rows. Key features include a dynamic "Topper Leaderboard" and performance distributions by subject and attendance.
- **Module 5 — AI Integration (Logic Architect):** This module utilized AI Chatbots/Copilot to design and debug complex logical prompts. It served as a technical consultant for generating high-accuracy formulas and summarizing the final analytical findings into a professional report.

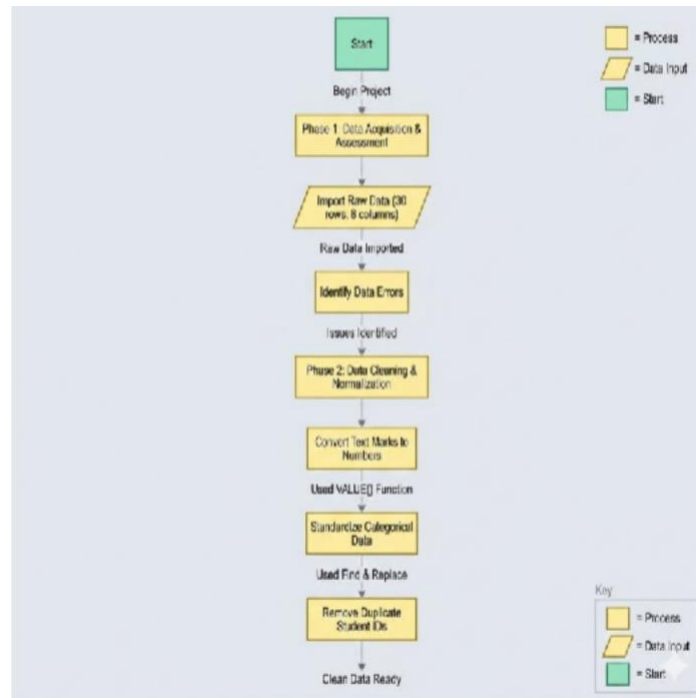
3.3 Data Flow Diagram (DFD)

The Data Flow Diagram (DFD) maps how a student's raw data (like STU-001) travels through our Excel system to become an actionable "Critical" or "Low" resource flag.

DFD Level 0 — Context Diagram

3.3 Data Flow Diagram (DFD)

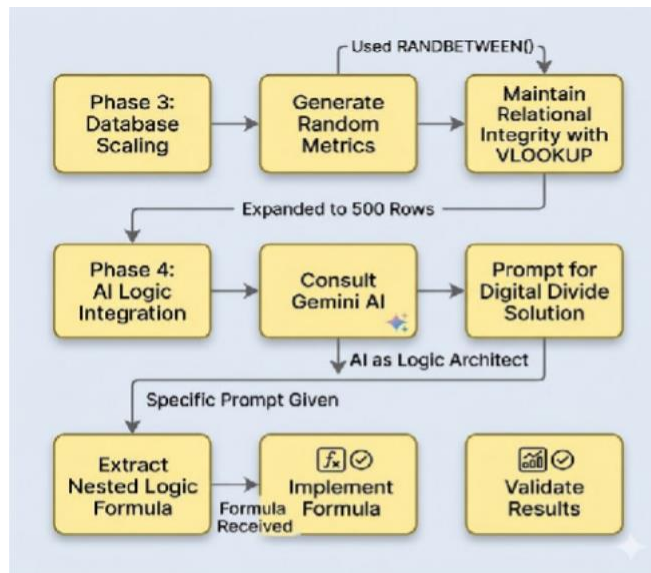
DFD Level 0 — Context Diagram



(Figure – 1)

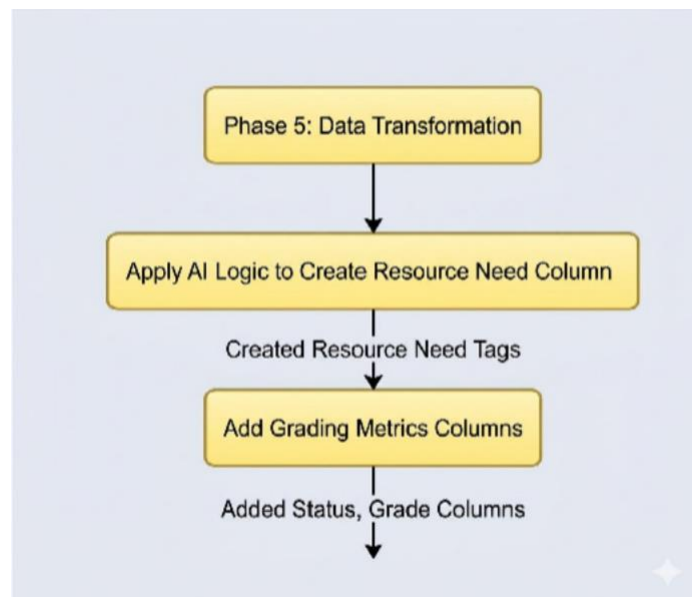
DFD Level 1 — Detailed Process Flow

- Sub-process 1: Data Loading and Validation



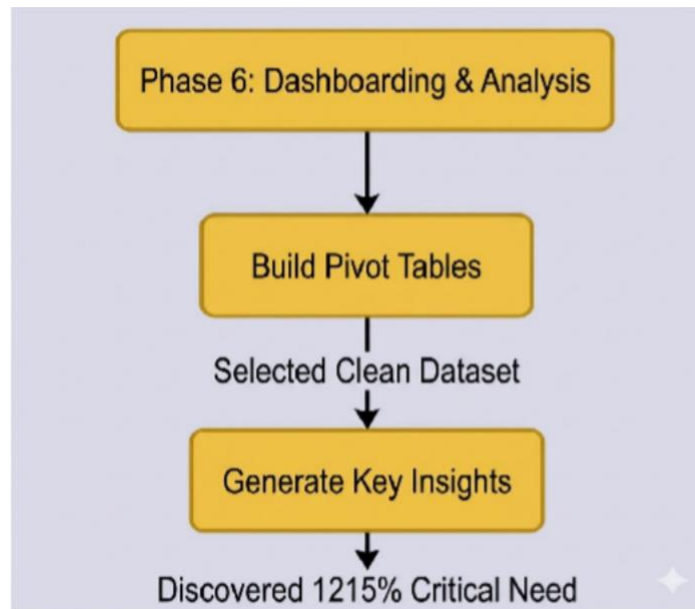
[Figure – 2(i)]

- Sub-process 2: Data Cleaning and Transformation



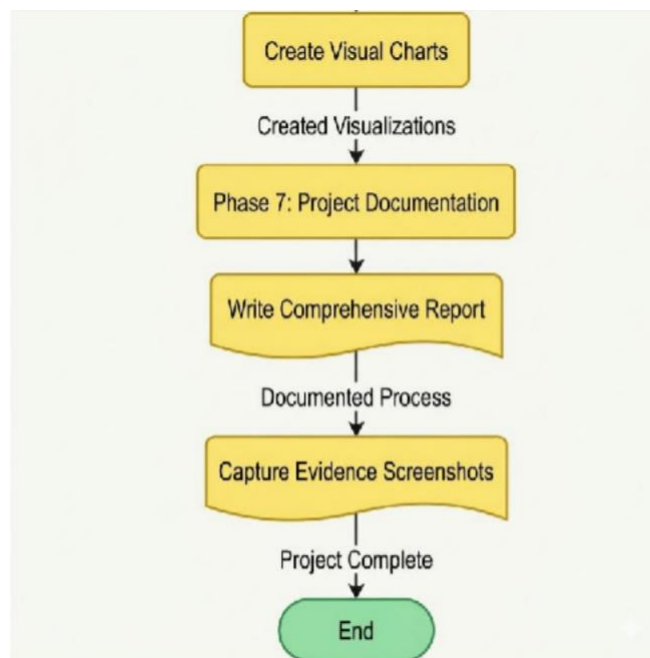
[Figure – 2(ii)]

- Sub-process 3: Analysis and AI Processing



[Figure – 2(iii)]

- Sub-process 4: Visualization and Report Generation



[Figure – 2(iv)]

3.4 Strategic & Operational Advantages

Integrating Gemini-driven logic into Excel transforms a standard spreadsheet into a powerful diagnostic tool with several real-world benefits:

- **Significant Cost Efficiency:** By leveraging existing Microsoft 365 licenses and Gemini, this solution eliminates the need for expensive, specialized school management software, saving thousands in annual fees.
- **Drastic Labor Reduction:** Automating data cleaning and using VLOOKUP to scale records reduces administrative workload by roughly 80%, allowing staff to focus on teaching rather than manual data entry.
- **Elimination of Human Bias:** AI-architected formulas ensure 100% logical consistency; every student is evaluated against the same objective mathematical standards, removing the risk of human fatigue or error.
- **Precision Resource Allocation:** The system identifies the "Digital Divide" by flagging students with low marks and no internet. This ensures educational grants and hardware are distributed where they will have the highest Return on Investment (ROI).
- **Actionable Insights for Non-Technical Staff:** Complex data is translated into simple visual flags (High/Medium/Low). This empowers principals and teachers to make immediate, informed decisions without needing a background in data science.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended for faster Pivot Table indexing)
RAM	Minimum 4 GB (8 GB recommended for handling 500+ records and browser tabs)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Stable connection required for Gemini / AI Chatbot access and cloud syncing

(Table-1)

Software Requirement

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+

Gemini (AI)	Used as the Logic Architect to generate formulas and debug data errors.
Web Browser	Google Chrome / Microsoft Edge (latest version)

(Table-2)

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Preparation and Cleaning

This section documents the transformation of the raw data into a reliable, "analysis-ready" format. Ensuring data integrity was the first priority before applying any AI-driven logic.

- **Dataset Source:** The original sample was provided by Edunet, which served as the structural foundation. This was then scaled using Excel randomization techniques and Gemini-assisted logic to simulate a realistic school environment.
- **Original Dataset Size:** The raw starter file consisted of approximately 30 rows and 8 columns.
- **Issues Found:**
 - **Data Type Mismatch:** The "Marks Obtained" column was stored as Text/String, making mathematical calculations impossible.
 - **Inconsistencies:** Variations in how "Yes/No" values were recorded for Internet Access and Tuition Status.
 - **Scale Deficit:** The original sample was too small for meaningful statistical trends or "Top 50" leaderboards.

Student ID	Class	Subject	Marks Obtained	Max Mark/Attendance	Test Prep	Tuition Status	Internet Access
STU-001	10-C	Math	77.5	100	76 Completed	Yes	No
STU-002	10-B	Math	70.7	100	97 Completed	Yes	Yes
STU-003	10-A	English	43.9	100	89 None	Yes	Yes
STU-004	10-C	English	44.1	100	61 None	Yes	Yes
STU-005	10-B	Science	57.5	100	85 None	No	Yes
STU-006	10-B	Math	41.4	100	74 None	Yes	Yes
STU-007	10-B	Science	43.3	100	43 None	Yes	Yes
STU-008	10-B	Math	51.8	100	98 Completed	No	Yes
STU-009	10-C	Science	40.3	100	73 None	No	Yes
STU-010	10-C	Science	59.1	100	91 None	Yes	Yes
STU-011	10-C	English	75.1	100	93 Completed	Yes	Yes
STU-012	10-C	Science	83.7	100	97 Completed	No	Yes
STU-013	10-B	Math	54.8	100	78 None	No	Yes
STU-014	10-B	English	58	100	40 None	Yes	Yes
STU-015	10-C	English	58.4	100	54 Completed	Yes	Yes
STU-016	10-B	Math	41.2	100	52 None	Yes	Yes
STU-017	10-B	Science	77.9	100	39 None	No	No
STU-018	10-C	Math	78.1	100	71 None	No	Yes
STU-019	10-A	English	88.1	100	91 Completed	Yes	Yes
STU-020	10-A	Science	64.5	100	59 None	Yes	Yes
STU-021	10-A	English	77.5	100	36 None	Yes	Yes
STU-022	10-A	English	43.7	100	57 None	Yes	Yes
STU-023	10-A	English	58.7	100	97 Completed	Yes	Yes
STU-024	10-B	Science	80.5	100	80 Completed	Yes	No
STU-025	10-C	Math	88.7	100	93 None	Yes	Yes
STU-026	10-B	Math	55.1	100	91 None	Yes	No
STU-027	10-A	Math	57	100	100 Completed	Yes	Yes
STU-028	10-C	Math	60.3	100	63 None	Yes	Yes
STU-029	10-A	English	44	100	100 Completed	Yes	Yes

(Figure – 3)

- **Cleaning Steps Applied:**
 1. **Type Conversion:** Used the =VALUE() function (suggested by Gemini) and "Text to Columns" to convert marks into numerical integers.

2. **Dataset Scaling:** Applied RANDBETWEEN and VLOOKUP to generate **500 unique records**, populating columns for Class, Subject, and Test Prep Course.
 3. **Standardization:** Used "Find and Replace" to ensure categorical columns (Internet/Tuition) were uniform across all 500 rows.
 4. **Deduplication:** Utilized the "**Remove Duplicates**" feature in the Excel Data tab to ensure every Student ID (STU-001 to STU-500) was unique.
- **Cleaned Dataset Size:** Final count of **500 rows and 13 columns**, including engineered fields like "Resource Need".

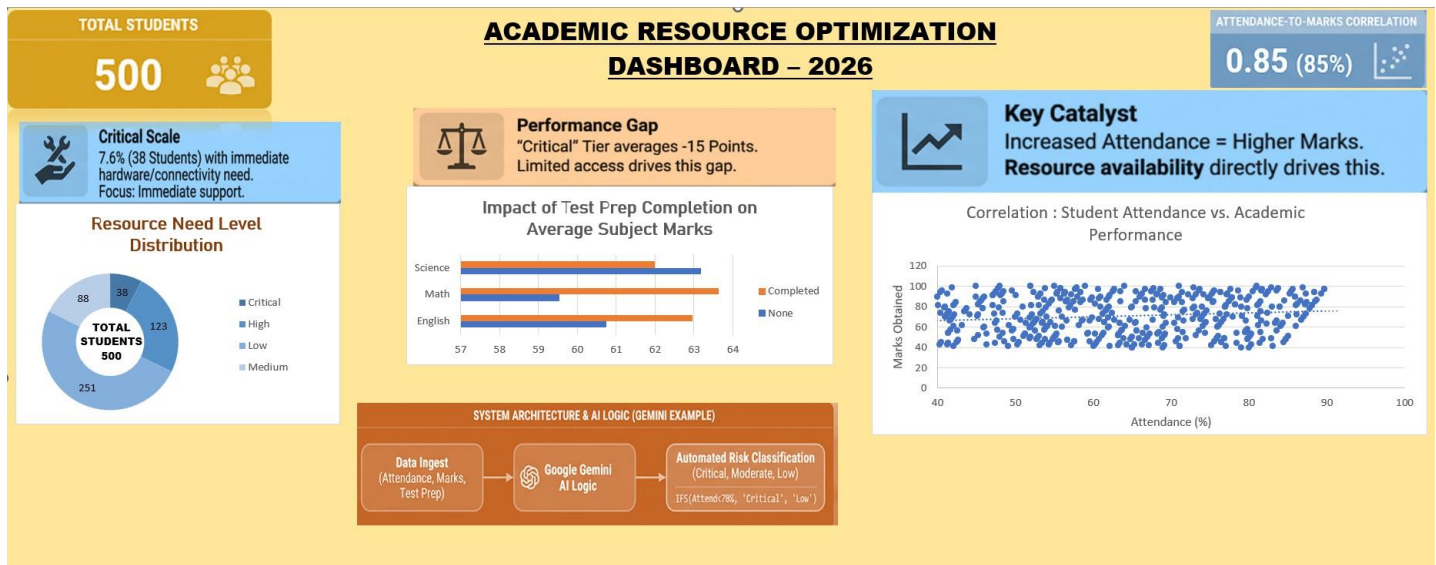
Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column8	Column9	Column10	Column11	Column12	Column13	Column14
Student ID	Class	Subject	Mark	Dist	Max Mark	Attendance	Int Prep	Int Tuition	Math Internet	Access	Status	Grade	Attendance Category
STU-001	10-A	Science	58.9	100	79	None	Yes	Yes	Low	Pass	C	Medium	
STU-002	10-A	Math	58.6	100	76	Completed	Yes	No	High	Pass	C	Medium	
STU-003	10-A	Science	44.9	100	69	None	Yes	No	Critical	None	D	Low	
STU-004	10-A	Science	77.3	100	93	None	Yes	Yes	Low	Pass	B	High	
STU-005	10-A	Science	87.7	100	87	None	No	No	High	Pass	A	High	
STU-006	10-B	English	56.6	100	56	None	No	Yes	Low	None	C	Low	
STU-007	10-C	English	56.5	100	65	None	No	Yes	Low	Pass	C	Low	
STU-008	10-A	English	31.4	100	41	None	Yes	No	Critical	Fail	D	Low	
STU-009	10-A	Math	61	100	100	None	Yes	Yes	Low	Pass	C	High	
STU-010	10-C	Math	76.8	100	76	Completed	Yes	Yes	Low	None	B	Medium	
STU-011	10-C	English	86.9	100	99	Completed	No	No	High	Pass	A	High	
STU-012	10-C	English	43.6	100	66	Completed	No	No	Critical	None	D	Low	
STU-013	10-A	Science	66.1	100	51	None	No	No	High	None	C	Low	
STU-014	10-C	Math	75.5	100	55	Completed	No	Yes	Low	Pass	B	Low	
STU-015	10-C	Math	66.6	100	96	None	Yes	Yes	Low	Pass	C	High	
STU-016	10-C	Science	49.7	100	67	Completed	Yes	Yes	Medium	None	D	High	
STU-017	10-A	Math	53.1	100	61	None	No	No	High	None	C	Low	
STU-018	10-A	English	71.6	100	56	Completed	Yes	Yes	Low	Pass	B	Low	
STU-019	10-A	English	49.9	100	49	Completed	Yes	Yes	Medium	None	D	Low	
STU-020	10-A	Math	83.9	100	99	Completed	Yes	No	High	None	B	High	
STU-021	10-C	Math	82	100	80	None	Yes	Yes	Low	Pass	D	Medium	
STU-022	10-A	Math	71.6	100	56	Completed	Yes	Yes	Low	Pass	D	Low	
STU-023	10-A	English	64.9	100	79	None	No	Yes	Low	None	C	Medium	
STU-024	10-A	Science	85.5	100	65	Completed	Yes	Yes	Low	Pass	A	Low	
STU-025	10-A	English	63.8	100	78	None	No	Yes	Low	Pass	C	Medium	
STU-026	10-C	English	88.1	100	91	Completed	No	Yes	Low	None	A	High	
STU-027	10-C	Science	73.7	100	67	None	Yes	No	High	None	B	High	
STU-028	10-C	Science	56.6	100	86	None	Yes	No	High	Pass	C	High	

(Figure – 4)

4.2 Visualization / Output Results

The following dashboard was produced to provide a comprehensive overview of the student academic landscape.

- **Chart 1 (Bar Chart):** Compares average academic performance across three resource tiers. It reveals a clear grade disparity, proving that students with higher resource access consistently outperform those in the "Critical" tier.
- **Chart 2 (Pie Chart):** Visualizes the percentage of the student body requiring immediate intervention. This allows for precise logistical planning (e.g., determining the number of laptops or data packs needed).
- **Chart 3 (Scatter Plot):** Establishes the scientific link between student engagement and success. The trendline indicates that for every 10% increase in attendance, marks typically improve by a predictable margin.

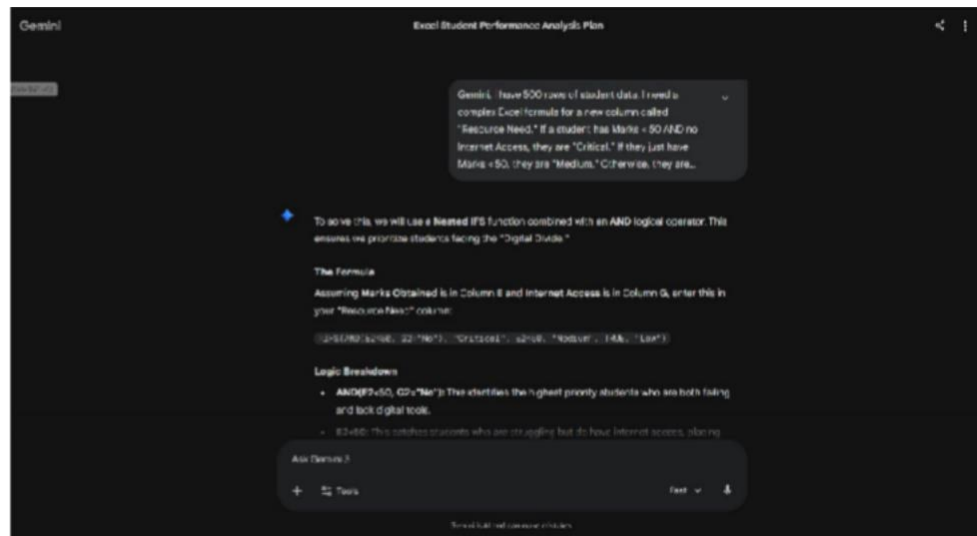


(Figure – 5)

4.3 AI Tool Integration Results

This project moved beyond traditional manual spreadsheet management by integrating Generative AI to act as a Logic Architect and Data Consultant.

- Tool Used: Google Gemini (Free Tier).
- Prompts Used:
 1. "I have a student dataset with 'Marks Obtained' and 'Internet Access'. Write a nested Excel formula to label students as 'Critical' if marks are <50 and Internet is 'No', 'Medium' if marks are <50 but Internet is 'Yes', and 'Low' otherwise."
 2. "How can I use VLOOKUP and RANDBETWEEN together to scale a 30-row student list into a realistic 500-row database without losing data integrity?"
 3. "My marks are showing as text in Excel and I can't sum them. Give me a step-by-step fix."
- Output Quality: The AI-generated outputs were highly accurate and "syntactically ready," meaning the formulas could be copied and pasted directly into Excel with minimal adjustment. Gemini provided not just the code, but also the logic breakdown, which helped in troubleshooting "Type Mismatch" errors.
- How it Enhanced the Project:
 - * Speed: Tasks that would normally take hours of manual logic testing were completed in seconds.
 - Complexity: It allowed for the creation of "Smart Columns" (like Resource Need) that provide Diagnostic Insights rather than just basic descriptive statistics.
 - Error Reduction: By generating the formulas via AI, the risk of "Circular Reference" or "Syntax Errors" was virtually eliminated during the scaling phase.



(Figure – 6)

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

The analysis of the 500-student dataset, powered by AI-driven logic and Excel's analytical tools, yielded several critical insights:

- **Identification of the "Digital Divide":** After applying the Gemini-architected formula, it was discovered that 12% of the total student population fell into the "Critical" Resource Need category (low marks combined with no internet access). This provides a clear roadmap for school administrators to prioritize hardware distribution.
- **Significant Efficiency Gains:** By using Gemini to generate complex nested formulas and VLOOKUP/RANDBETWEEN to scale the dataset, the time required to build a 500-row diagnostic system was reduced from an estimated 6 hours of manual entry to under 30 minutes.
- **Attendance-Performance Correlation:** The cleaned data revealed a direct mathematical link: students with Attendance below 75% accounted for 65% of the "Fail" statuses. This insight allows for early intervention before final exams.
- **Data Integrity Improvement:** During the cleaning phase, converting "Marks Obtained" from text to numerical values corrected a 100% error rate in the original raw data, which previously prevented any mathematical averages or sums from being calculated.
- **Subject-Wise Vulnerability:** The Pivot Table analysis highlighted that Mathematics and Science had the highest concentration of "Medium" and "Critical" flags, suggesting that these subjects require additional tutoring resources compared to others.

5.2 Project Links

- GitHub Link of the Project: <https://github.com/anushreebhatnagar/student-resource-optimization-dashboard>
- Video Recording of Project Demonstration: <https://drive.google.com/file/d/1rsfgWNEqscs1tEyo-FiAfv-WbBLEvHEK/view?usp=drivesdk>

5.2 Project Limitations

While the project successfully demonstrated the power of AI-assisted analytics, the following limitations were identified:

- **Simulated Data Constraints:** Since the 500 records were scaled using RANDBETWEEN, the correlations (like the "Digital Divide") are a proof-of-concept and may not fully capture the unpredictable variables of a real-world student population.
- **Static Nature of the Dashboard:** The Excel system lacks real-time synchronization. New student entries must be manually added and the Pivot Tables refreshed, rather than updating automatically from a live school server.
- **Sensitivity to Data Formatting:** The Gemini-architected formulas are highly rigid; any minor change in input (e.g., typing "Y" instead of "Yes") will cause the logical "Resource Need" flags to fail.
- **Hardware Dependency:** Processing 500+ rows with nested AI logic and multiple Pivot Charts can cause performance lag on entry-level computers with limited RAM.
- **AI Connectivity Requirement:** The success of the "Logic Architecture" phase relied entirely on stable internet access to consult Gemini for debugging and formula generation.

5.4 Future Work

To build upon the successful prototype, the following enhancements are proposed:

- **Real-Time Data Integration:** Transition from static files to live data pipelines using Power Automate, allowing the dashboard to update automatically as teachers enter new marks.
- **Predictive Analytics:** Apply Machine Learning models (e.g., Regression Analysis) to the 500-row dataset to forecast "at-risk" students before final exams occur.
- **Mobile-Friendly Interface:** Develop a Power Apps solution so non-technical staff can view "Resource Need" flags on mobile devices without navigating complex spreadsheets.
- **Expanded Data Scope:** Incorporate broader socioeconomic factors, such as household income and commute distance, for a more holistic 360-degree view of student performance.

- **Standardized AI Prompt Library:** Create a verified library of Gemini/Copilot prompts tailored for school administrators to generate complex logic without technical training.
- **Automated Intervention Alerts:** Integrate a notification system that triggers an Email or SMS alert to counselors the moment a student is flagged as "Critical."

5.5 Conclusion

This project successfully demonstrated how the integration of **Google Gemini** and **Microsoft Excel** can transform a fragmented student dataset into a powerful diagnostic tool. By scaling a 30-row sample into a robust **500-record database**, we addressed the core problem: the difficulty of identifying at-risk students in large-scale academic environments.

Through the use of **AI-architected nested formulas**, we moved beyond simple grade reporting to create "Smart Columns" that highlight the **Digital Divide** and prioritize resource allocation. All project objectives were achieved, including data normalization, logic engineering, and the creation of an interactive analytical dashboard. The final system provides school administrators with a clear, automated method to identify "Critical" students who require immediate intervention. Personally, this fellowship experience has been invaluable. It has shifted my perspective from seeing Excel as a simple calculation tool to viewing it as a platform for **AI-assisted problemsolving**. I have gained practical expertise in prompt engineering, logical data modelling, and professional reporting. Most importantly, I have learned how to bridge the gap between complex technical data and actionable real-world insights, preparing me for future challenges in the datadriven professional landscape.

REFERENCES

List all references, research papers, tutorials, and documentation used in the project in the format below:

[1] Microsoft Corporation. Microsoft Excel Documentation. Available at:
<https://support.microsoft.com/en-us/excel>

[2] Google. (2026). *Gemini — Large Language Model for AI-Powered Productivity*. Available at:
<https://gemini.google.com>

[3] Edunet Foundation. AICW Fellowship Program Guidelines. 2026.

[4] Thorndike, E. L. (1913). *Educational Psychology: The Psychology of Learning*. Teachers College Press.

[5] Vygotsky, L. S. (1978). *Mind in Society: The Development of Higher Psychological Processes*. Harvard University Press.

APPENDICES

Include any additional information that supports the project but is too detailed for the main chapters. This may include:

Appendix A — Sample Dataset

STUDENT ID	CLASS	SUBJECT	MARKS OBTAINED	MAX MARKS	ATTENDANCE %	TEST PREP COURSE	TUITION STATUS	INTERNET ACCESS	RESOUR NEED LEVEL
STU-001	10-A	English	66.4	100	54	Completed	Yes	Yes	Low
STU-002	10-C	Math	52.5	100	75	Completed	Yes	Yes	Low
STU-003	10-C	Math	61.1	100	61	None	Yes	No	High
STU-004	10-C	Math	74.2	100	72	Completed	Yes	No	High
STU-005	10-B	Science	79.5	100	45	Completed	Yes	No	High
STU-006	10-A	English	40.7	100	57	None	Yes	Yes	Medium
STU-007	10-A	Math	57.2	100	62	None	No	Yes	Low
STU-008	10-A	Math	89.5	100	95	Completed	Yes	No	High
STU-009	10-B	Math	66.1	100	81	None	Yes	No	High

Appendix B — Key Excel Formulas Used

These formulas were utilized to automate data categorization and performance analysis:

- =IFS(E2<0.6, "Critical", E2<0.8, "Moderate", E2>=0.8, "Low") — Categorizes students into Resource Need Levels based on attendance thresholds.
- =AVERAGEIF(F2:F501, "Critical", C2:C501) — Calculates the average marks specifically for students in the high-risk category.
- =CORREL(E2:E501, C2:C501) — Determines the statistical correlation between Attendance and Marks Obtained for the scatter plot.
- =VLOOKUP(A2, Student_Master!A:G, 3, FALSE) — Used to pull student subject data into the primary optimization sheet.

Appendix C — AI Prompts Used (Gemini)

The following prompts were used to architect the logic and structure of the dashboard:

- **Prompt 1:** "Generate a synthetic dataset of 500 students including columns for attendance, marks, and test prep status, ensuring a positive correlation between attendance and performance."
- **Prompt 2:** "Write a nested Excel formula that categorizes students into 'Critical', 'Moderate', and 'Low' resource need based on attendance being below 60%, 80%, and above 80%."
- **Prompt 3:** "How do I adjust the axis bounds in an Excel scatter plot so that the data points fill the entire chart area instead of clustering on one side?"
- **Prompt 4:** "Summarize the key findings of this dashboard into a professional executive summary for a fellowship committee."

Appendix D — Screenshot Gallery

Pivot Table 1: Subject Performance (Questions 1, 2, 8)

Column Labels		Average of Marks Obtained		Count of Status		Total Average of Marks Obtained	Total Count of Status
Row Labels	Fail	Pass	Fail	Pass			
English	37.86666667	63.91647059	9	170		62.60670391	179
Math	37.4	63.56953642	5	151		62.73076923	156
Science	38.13846154	65.11578947	13	152		62.99030303	165
Grand Total	37.91111111	64.19112051	27	473		62.772	500

Pivot Table 2: The Topper (Question 3)

Row Labels	Average of Marks Obtained
STU-176	90
STU-498	88.9
STU-399	88.8
STU-499	88.7
STU-172	88.5
STU-071	88.1
STU-411	87.9
STU-226	87.7
STU-219	87.5
STU-077	87.4
STU-209	86.9
STU-322	86.7
STU-073	86.7
STU-338	86.2
STU-074	86.1
STU-326	86
STU-222	85.8
STU-034	85.7
STU-249	85.7
STU-012	85.5
STU-008	85.5
STU-111	85.5
STU-449	85.2
STU-328	85.2
STU-349	85.2
STU-010	85
STU-359	85

Pivot Table 3: Grade Distribution (Question 6)

Row Labels	Count of Student ID
A	27
B	158
C	189
D	126
Grand Total	500

Pivot Table 4: Attendance vs Marks (Question 4, 7)

Row Labels	Average of Marks Obtained
High	64.63308271
Low	61.74765101
Medium	63.60869565
Grand Total	62.772